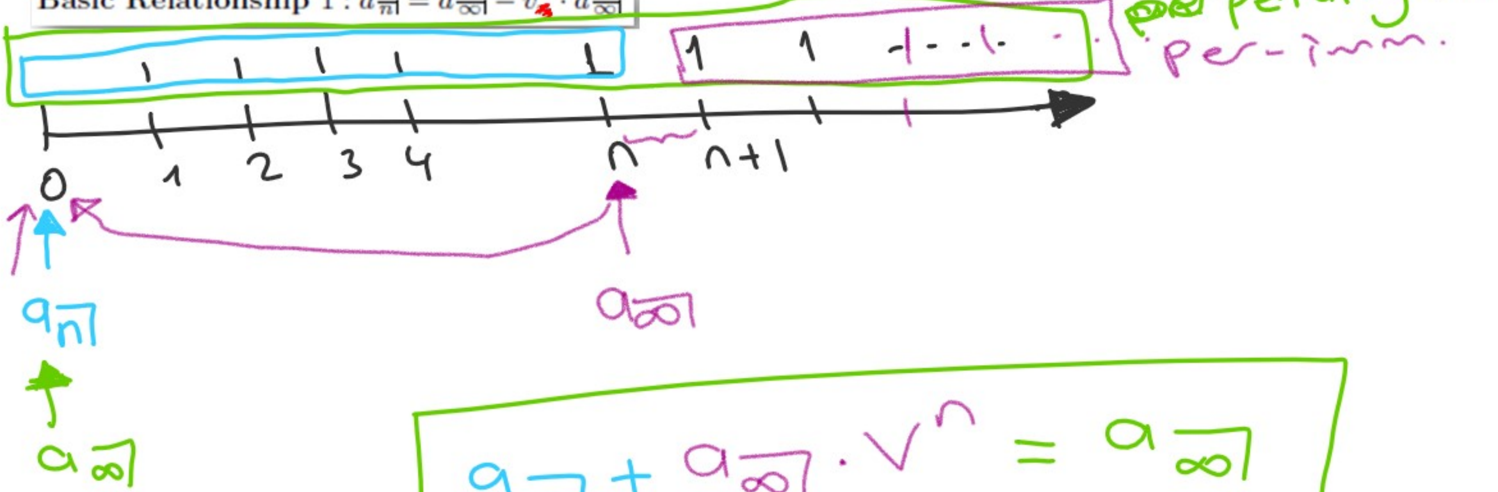
17.04.2020 Finance Math.

14 Nisan 2020 Salı 19:57

Basic Relationship 1 : $a_{\overline{n}|} = a_{\overline{\infty}|} - v^n \cdot a_{\overline{\infty}|}$



$$\boxed{a_{\overline{n}|} + a_{\overline{\infty}|} \cdot v^n = a_{\overline{\infty}|}}$$

2000 → year　　→ 2011

12. Payments of $100 per quarter are made from June 7, Z through December 7, Z + 11, inclusive. If the nominal rate of interest convertible quarterly is 6%:
   a) Find the present value on September 7, Z - 1. → case 1
   b) Find the current value on March 7, Z + 8. → case 3.
   c) Find the accumulated value on June 7, Z + 12. → case 2

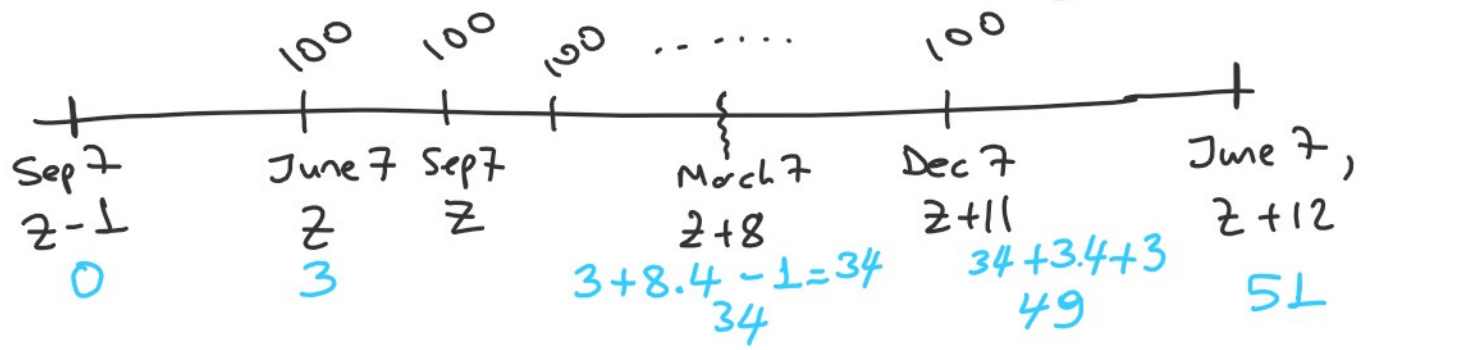
$i^{(4)} = 0.06$

6th June 3

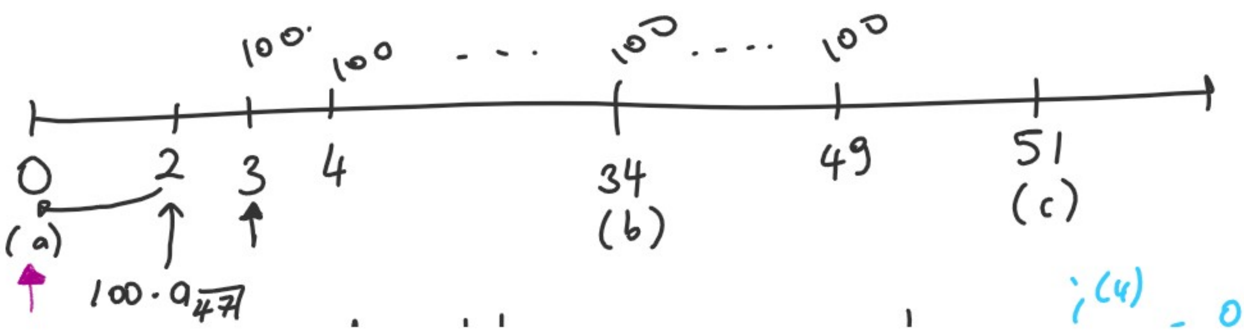
9th Sep. 0

12th Decem. 1

3th March. 2



J



$i^{(4)} = 0.06$

(a) $100 \cdot a_{\overline{47}|}$

number of the payments: $\frac{i^{(4)}}{4} = \frac{0.06}{4} = 0.015 = j$

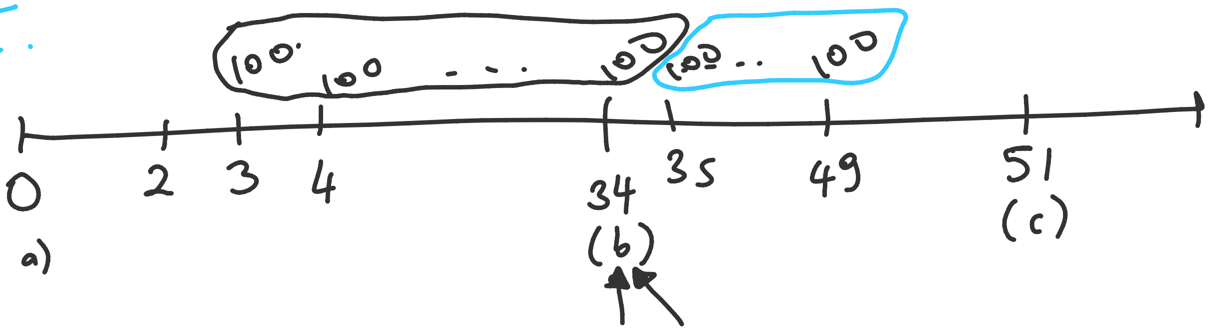
$$49 - 3 + 1 = 47$$

(a) => $100 \cdot a_{\overline{47}|j} \cdot (1+j)^{-2}$ or

$100 \cdot \ddot{a}_{\overline{47}|j} \cdot (1+j)^{-3}$.

$\begin{array}{r} 32 \\ +15 \\ \hline 47. \end{array}$

$34 - 3 + 1 = 32$ # $49 - 35 + 1 = 15$



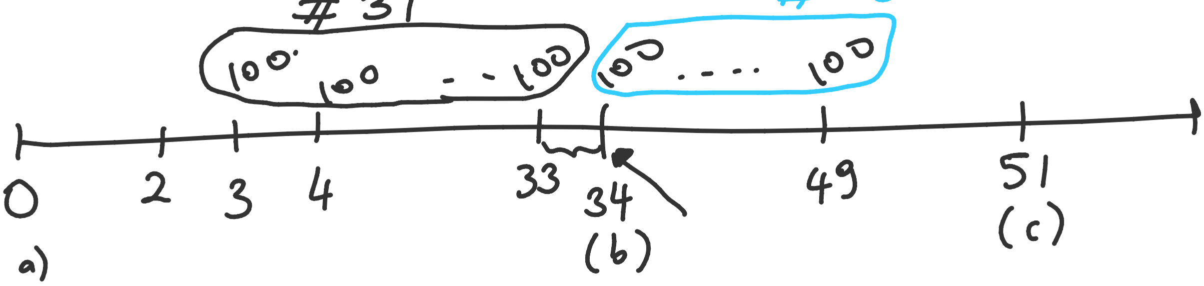
(b) ann-imm.

$$100 \cdot s_{\overline{32}|j} + 100 \cdot a_{\overline{15}|j}$$

or

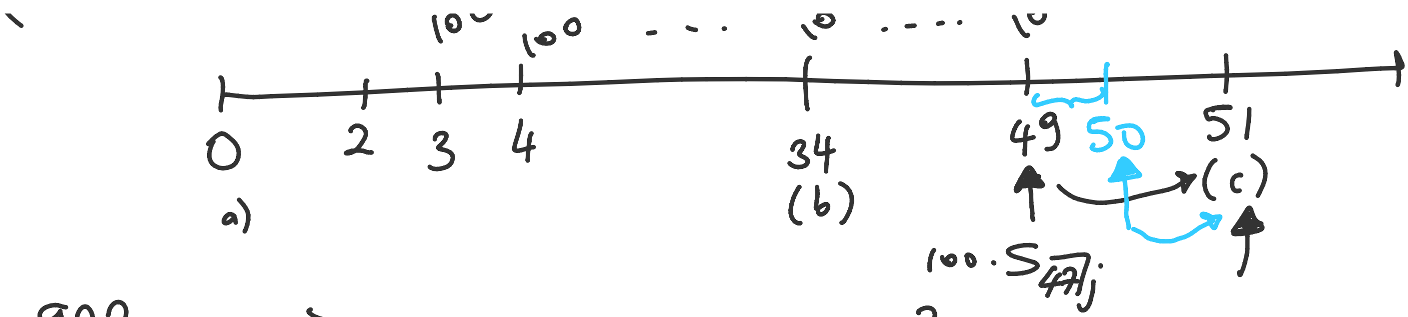
#31 #16

ann due



$$100 \cdot \ddot{s}_{\overline{31}|j} + 100 \cdot \ddot{a}_{\overline{16}|j}$$

(c) #47 payments.



ann imm $\Longrightarrow 100 \cdot s_{\overline{47}|j} \cdot (1+j)^2$

or

ann due $\Longrightarrow 100 \cdot \ddot{s}_{\overline{47}|j} \cdot (1+j)^1$

12. Payments of \$100 per quarter are made from June 7, Z through December 7, Z + 11, inclusive. If the nominal rate of interest convertible quarterly is 6%:
    a) Find the present value on September 7, Z - 1.
    b) Find the current value on March 7, Z + 8.
    c) Find the accumulated value on June 7, Z + 12.

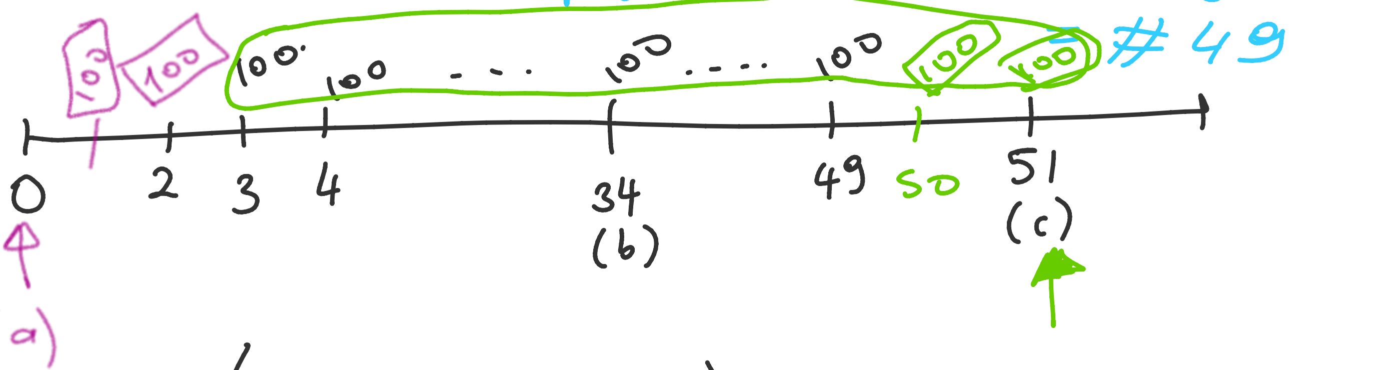
12. We will call September 7, $z-1$ $t=0$
so that March 7, $z+8$ is $t=34$
and June 7, $z+12$ is $t=51$
where time t is measured in quarters. Payments are made at $t=3$ through $t=49$, inclusive. The quarterly rate of interest is $j=.06/4=.015$.

(a) $\text{PV}=100(a_{\overline{49}|}-a_{\overline{2}|})=100(34.5247-1.9559)=\$3256.88.$

(b) $\text{CV}=100(s_{\overline{32}|}+a_{\overline{15}|})=100(40.6883+13.3432)=\$5403.15.$

(c) $\text{AV}=100(s_{\overline{49}|}-s_{\overline{2}|})=100(71.6087-2.0150)=\$6959.37.$

47 payments + 2 imaginary



$100 \cdot (a_{\overline{49}|j} - a_{\overline{2}|j})$

(c) $100 \cdot (s_{\overline{49}|j} - s_{\overline{2}|j})$

OR

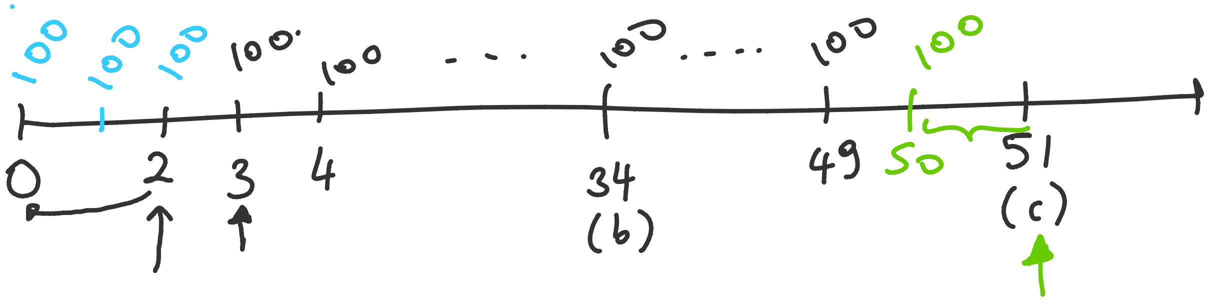
ann

OR

annu
due.

$47 + 1 = 48$

\# 50



$$(a) \quad 100 \cdot \left(\ddot{a}_{\overline{50}|j} - \ddot{a}_{\overline{3}|j}\right)$$

$$(c) \quad 100 \left(\ddot{s}_{\overline{48}|j} - \ddot{s}_{\overline{1}|j}\right)$$